

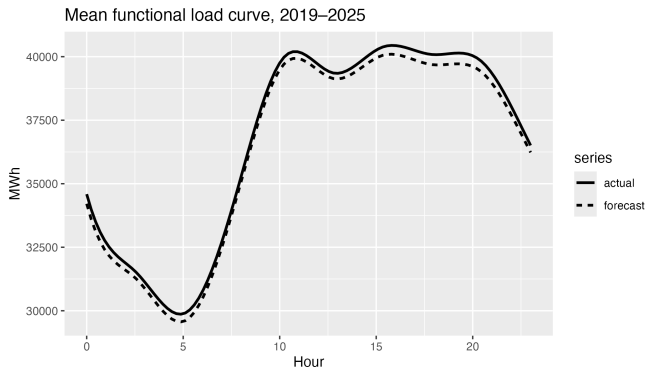
Functional Forecasting of Turkish Electricity Load Curves

MRes Thesis Proposal

Object	Daily load curve $Y_d(h)$, $h = 0, \dots, 23$ (MWh/hour)
Benchmark	Official day-ahead forecast published by TEİAŞ / EPIAS
Evaluation	Rolling-origin, 2023–2025, 1,096 test days, 26,304 hourly errors
Main test	Diebold–Mariano (pooled and hour-by-hour)

The Forecasting Object: A Daily Load Curve

$$Y_d(h), \quad h = 0, \dots, 23$$



- ▶ A single **curve**, not 24 scalars
- ▶ Dual-peak shape is structurally stable across all days
- ▶ Solid = actual; dashed = official forecast
- ▶ **Goal:** forecast tomorrow's full 24-hour profile

The official forecast (dashed) lies *systematically below* the actual — persistent underprediction at every hour.

The Load Curve Is an Economic Story

Segment	Hours	Economic driver	vs. official	
Overnight trough	0–5	Factories + infra; stable, weekly-seasonal	Wins 60%	40–
Morning ramp	6–8	Demand doubles; ramp timing temperature-driven	Loses (up to 24%)	
Midday plateau	9–16	Industrial & commercial peak; weekday-driven	Wins 35%	20–
Evening surge	17–19	Residential peak; AC/heating at max; largest official bias	Mixed	
Night wind-down	20–23	Smooth decline toward trough	Wins 20%	17–

Why functional analysis fits:

These 24 observations are one coherent trajectory shaped by overlapping behavioral rhythms.

The morning ramp, midday plateau, and evening peak are *features of the whole curve* — not of individual hours. FDA works with these shape features directly via eigenfunctions and derivative summaries.

Red rows: transition segments where demand changes fastest and forecasting is hardest.

Why Accuracy Matters

Key users (post-2013 liberalisation):

- ▶ **TEİAŞ** (transmission system operator)
unit commitment, reserve scheduling
- ▶ **Generators**
day-ahead market bids on EPIAS
- ▶ **Large consumers**
real-time imbalance penalties

Why does the official underpredict?

Most likely: **model staleness**. Official systems require regulatory approval to update; a model calibrated on older demand patterns underpredicts during structural demand growth.

Our rolling-origin re-trains before every step
— 2024 bias: **34 MWh** vs. official **585**

Systematic underprediction: official TEİAŞ forecast

Year	Official bias (MWh/hour)	Our bias (MWh/hour)
2023	+344	+176
2024	+585	+34
2025	+472	+98
Average	+467	+103

Positive bias = forecast < actual (underprediction).

Research Questions

1. **Representation.** Can B-spline smoothing and FPCA represent Turkey's daily electricity load curves with negligible information loss? How many functional principal components suffice?
2. **Forecasting accuracy.** Does a functional VAR model augmented with holiday and derivative features statistically outperform the official TEİAŞ day-ahead forecast? Evaluated by Diebold-Mariano over 1,096 hold-out days, 2023–2025.
3. **Nonlinearity.** Within the FPCA framework, does replacing OLS with a nonlinear learner (Random Forest) improve forecast accuracy? And does bypassing the functional representation altogether — forecasting each hour directly — offer further gains? Preliminary evidence from a one-year subsample raises this as an open question for the thesis.

Literature Review

Systematic review in progress.

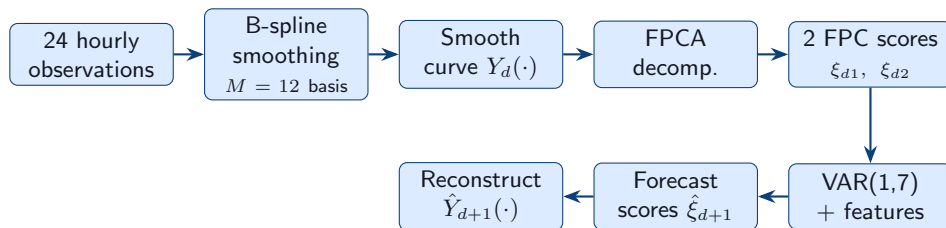
Established so far:

- ▶ No FDA-based study of Turkish electricity demand exists in the literature.

Areas to be covered in the full review:

- ▶ Functional Data Analysis foundations and forecasting methods (Ramsay & Silverman 2005; Kokoszka & Reimherr 2017; Hyndman & Ullah 2007)
- ▶ FDA applied to electricity load curves and energy demand
- ▶ Short-term load forecasting: classical (ARIMA) and machine learning approaches
- ▶ Turkish electricity market structure post-2013 liberalisation
- ▶ Residual correction and error correction in forecast combination (Xie & Hong 2016)

From 24 Numbers to a Curve: The FDA Pipeline

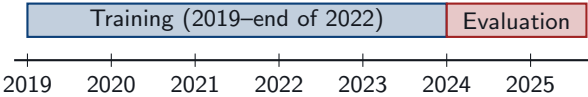


B-splines ($M = 12$)
preserves shape; enables
differentiation of the curve

FPCA ($K = 2$)
2 scores carry 97.3% of all curve
variation

VAR(1,7) by OLS
daily + weekly score dynamics

Data



Rolling-origin: model re-estimated before every forecast day using only past data. Expanding window — appropriate because eigenfunction shapes are stable across sub-periods.

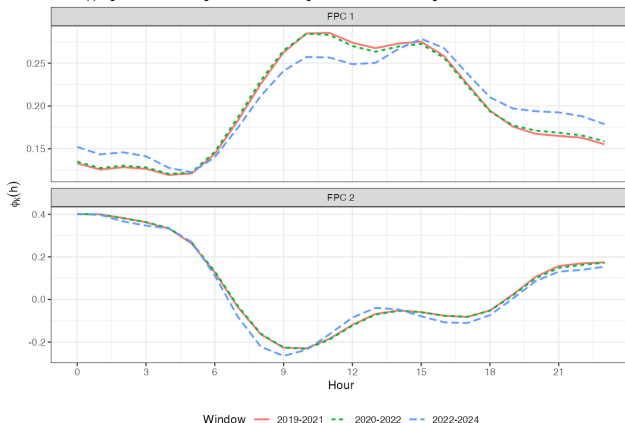
Series	Source
Hourly load (MWh)	EPIAS
Official forecast	TEİAŞ / EPIAS
Holiday calendar	Official gazette

Item	Value
Frequency	Hourly
Coverage	2019–2025
Training curves	≈1,460 days
Evaluation curves	1,096 days
Hourly errors eval	26,304

FPCA: Two Scores Capture the Curve

FPCA eigenfunctions across sub-periods

Overlapping lines = stable eigenfunctions; divergence = structural change



FPC	Variance	Cumulative
1	93.4%	93.4%
2	3.9%	97.3%
3	1.1%	98.4%
4	1.0%	99.4%

FPC 1 — Level

Nearly flat $\Rightarrow$ shifts the whole curve up or down uniformly.

FPC 2 — Shape contrast

Positive overnight & evening, negative daytime $\Rightarrow$ peak-vs-plateau contrast.

Stability: sub-period inner products > 0.99

$\Rightarrow$ static FPCA justified empirically

Robustness: $K = 4$ adds $< 1\%$ variance:

Score Forecasting Model

Core dynamics (VAR):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

- ▶ A_1 : yesterday's level and shape
- ▶ A_7 : same-day-last-week seasonality
- ▶ z_d : feature vector, *all predetermined*

Reconstruction:

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\phi}_1(h) \hat{\xi}_{d1} + \hat{\phi}_2(h) \hat{\xi}_{d2}$$

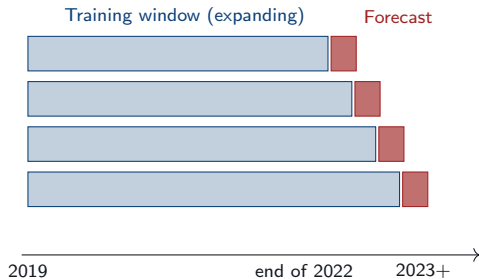
Estimated by OLS, equation-by-equation.

Feature blocks in z_d :

Block	Content
Calendar	Weekend flag, annual sine/cosine
Holiday	Fixed holidays, Ramadan, Bayram, bridge days
Load shape	Yesterday's mean, peak, trough, range
Rolling avg.	7-day, 28-day load averages
Derivative	Morning ramp (hr 6–8), evening ramp, slope volatility

Holiday and **derivative** blocks are critical: without them the model does not beat the official.

Evaluation: Rolling-Origin Design and DM Test



FPCA and VAR re-estimated at every step.
Only data dated $< d$ used to forecast day d .
Mirrors real operational deployment: no information leakage.

Diebold-Mariano test

Loss differential per observation:

$$d_t = |e_t^{\text{ours}}| - |e_t^{\text{official}}|$$

$H_1: \mathbb{E}[d_t] < 0$ (our model has lower loss)

HAC t -test on $\bar{d}$ (Newey-West long-run variance)

+ Harvey–Leybourne–Newbold small-sample correction.

Run **pooled** (26,304 obs.)

+ **hour-by-hour** (24 tests).

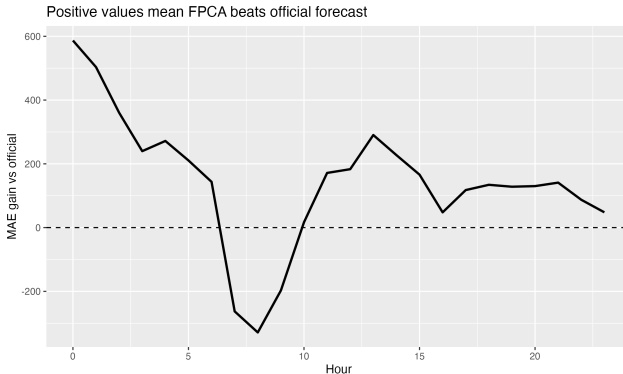
A lower MAE alone is not sufficient evidence.

Preliminary Results: Building the Model Step by Step

Model	MAE	MAPE	vs. official	DM stat	
Seasonal naïve Y_{d-7}	1912	5.13%	-57.6%	+35.2	—
Official (TEİAŞ)	1214	3.15%	ref.	—	
FPCA VAR(1,7) only	1443	3.76%	-18.9%	+22.9	
+ calendar / ramp	1238	3.26%	-2.1%	+2.6	
+ holiday / load-shape	1001	2.63%	+17.5%	-24.9	***
+ derivative features	980	2.56%	+19.2%	-27.4	***
+ residual correction	653	1.70%	+46.2%	-71.2	***

*** $p < 0.001$ vs. official (pooled DM, 26,304 obs.). Positive DM stat = worse than official; negative = better. MAE in MWh/hour. Evaluation: 2023–2025.

Where the Model Wins and Where It Loses



Positive = model beats official
(MAE, MWh/hour)

Wins: hours 0–5 and 10–23

Overnight trough and daytime plateau
— stable, weekly-seasonal behavior
well captured by VAR lags

Loses: hours 6–9

Morning ramp. Demand doubles in 2
hours; ramp timing is
temperature-driven. At hour 8, our
MAPE loss is **24%**.

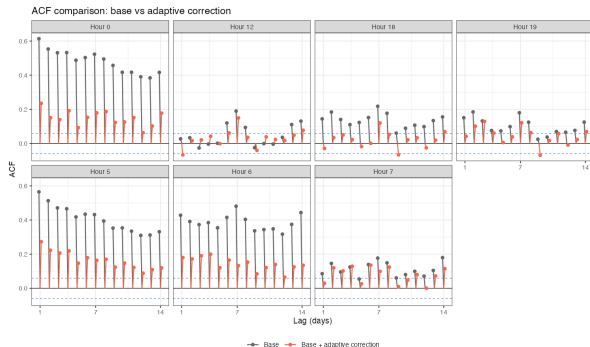
DM significant at 18 of 24 hours (5%
level).

Persistent, autocorrelated ramp-hour
errors motivate the residual
correction.

Residual Correction: Exploiting Persistent Errors

Motivation: errors are serially correlated

Ljung-Box rejects $H_0: \rho_1 = \dots = \rho_{14} = 0$ at all 24 hours. LB stat up to 7,227 at hour 6.



$$\hat{Y}_d^{\text{res}}(h) = \hat{Y}_d^{\text{base}}(h) + \lambda_d \hat{r}_d(h)$$

$\hat{r}_d(h)$: correction curve from recent past residuals

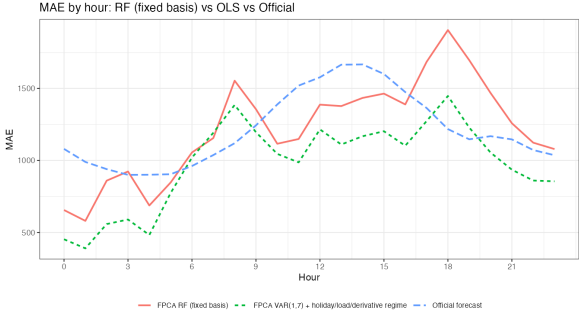
λ_d : rolling CV on strictly pre- d data
— no leakage

Result:

► +46.2% MAE vs. official (DM stat -71.2)

► Wins at all 24 hours including

Nonlinear Extensions: What the RF Results Suggest



Within FPCA structure — full 2023–2025

OLS replaced by RF in score equation; same basis, features, rolling-origin

Model	MAE	vs. off.	DM
Official	1214	ref.	—
FPCA-RF	1217	−0.3%	n.s.
FPCA-OLS	980	+19.2%	***

Within FPCA, OLS outperforms RF — the linear score equation is an effective regulariser.

Bypassing FPCA (Direct RF, 2025 only):

Predicts each hour directly from raw features. MAE 718 vs. official 1052 (+31.8%), FPCA-OLS 910 (+13.5%).

This is a one-year preliminary result — not directly comparable to the 2023–2025 evaluation — but it suggests that fully nonlinear architectures that bypass the functional compression may offer further gains.

Extending this comparison to the full three-year

Proposal Summary: Preliminary Findings and Open Questions

Preliminary findings (proposal phase)

- ▶ FPCA with $K = 2$ eigenfunctions represents load curves with 97.3% variance retention; eigenfunctions stable over time
- ▶ FPCA-VAR with holiday and derivative features achieves +19.2% MAE vs. official (DM $p \approx 10^{-163}$, 2023–2025)
- ▶ Residual correction brings gains to +46.2%, winning at all 24 hours including ramp hours
- ▶ Within the FPCA structure, OLS outperforms RF, suggesting the linear score equation is well-specified for this functional representation

Open questions for the thesis

- ▶ Does bypassing FPCA entirely (Direct RF, other nonlinear architectures) offer sustained gains over the full 2023–2025 evaluation? The 2025 subsample (+31.8%) is encouraging but based on a single year.
- ▶ Is the FPCA functional compression a limitation or a feature when paired with nonlinear score models?
- ▶ Can the ramp-hour gap be closed without external weather inputs?
- ▶ What are the gains from probabilistic forecasting (prediction intervals)?

Thesis Road Map

Proposal-phase analysis (done)

- ✓ Functional representation (B-splines, FPCA)
- ✓ Rolling-origin evaluation framework
- ✓ VAR(1,7) + feature blocks (holiday, derivative)
- ✓ Residual correction step
- ✓ DM testing (pooled + hourly)
- ✓ Eigenfunction stability; $K = 4$ check
- ✓ RF comparison within FPCA (score linearity)
- ✓ Direct RF — preliminary, 2025 subsample

Thesis objectives

- ☐ Systematic literature review
- ☐ Extend Direct RF to full 2023–2025 evaluation
- ☐ Investigate nonlinear functional architectures (RF, gradient boosting, neural nets on scores)
- ☐ Ramp-hour diagnostics and write-up
- ☐ Probabilistic forecasting (prediction intervals)
- ☐ Thesis write-up

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